

In-context unlearning: membership AUC under corruption

Stage 1 (AUC vs Var(eps) sweep) and Stage 2 (learned corruption policy)

Working report — 30 July 2026

Summary. Stage 1 is complete and validated: the three corruption families from the brief are implemented, swept, and the measured AUC agrees with an independently derived closed form to 4×10^{-4} . Four control configurations were run; two are exact nulls and both pass. The headline finding is negative and reproducible: the membership signal at zero corruption does *not* vanish when the group spectra are equalised, so it is not an artefact of one group being intrinsically harder. Stage 2 is implemented and demonstrated on a synthetic instance. Its main result is that the score-function (REINFORCE) estimator specified in the brief is not needed for the parameterisation the brief itself writes down — the objective is available in closed form — and that the objective as written minimises the wrong quantity.

1. The brief

The handwritten notes specify two stages.

Stage 1. A model reads a query x_q , a retain set D_r , and a corrupted forget set in context. Targets are linear, $y_f = w^T x_f$ and $y_r = w^T x_r$. The forget point is corrupted three ways — $(x_f, y_f + \text{eps})$, $(x_f + \text{eps}, y_f)$, and both — and the deliverable is membership AUC plotted against $\text{Var}(\text{eps})$.

Stage 2 (“Generalize”). Replace the hand-designed corruption with a learned one, $p_{\text{theta}}(\tilde{x}_f, \tilde{y}_f \mid x_f, y_f)$, factorised as a Bernoulli(theta) label flip with the features passed through unchanged. Optimise $\min_{\text{theta}} \text{AUC}(\text{theta}) + \lambda \|\text{theta}\|$. Because AUC is not differentiable through the sampling, the notes derive a score-function gradient, $\text{grad } E[\text{AUC}] = E[\text{grad } \log Q_{\text{theta}} \cdot \text{AUC}]$.

2. Stage 1: implementation

The three corruption arms map one-to-one onto the codebase as C1 (label noise), C2 (input noise) and C3 (both), swept over 21 log-spaced points across four decades. Each sweep point is evaluated on frozen shadow ensembles — 512 shadow models per hypothesis, trained once and reused, since a prompt-time edit never touches training. Three training seeds are crossed with three probe seeds, giving nine independent estimates per point.

Beyond the brief, four things were added because they change what the numbers mean:

Matched-context comparison. Both hypotheses are scored on the *same* corrupted prompt. Scoring the retrain oracle on a clean prompt lets a distinguisher win by detecting that an edit happened at all, and that confound grows with the very axis being plotted.

A masking control. Per-shadow corruption noise inflates within-ensemble spread, which lowers AUC mechanically without removing anything. Repeating each point with one noise draw broadcast across shadows separates that from genuine removal.

A closed-form overlay. The model is linear in the context, so the effect of Gaussian context noise on the read-out is available analytically. Drawing it under the measurements turns the figure into a standing correctness check.

Null controls. Two configurations give all three groups the same covariance spectrum *and* the same basis, making them one distribution. The full and oracle models then train on indistinguishable data, so AUC must be chance. If it is not, the two hypotheses are coupled somewhere in the code and nothing else is interpretable.

3. Stage 1: results

3.1 Baseline membership signal

AUC at zero corruption, on the sign-aligned residual. Values are means over three training seeds; the range is the spread of the three per-seed means.

configuration	groups	ATTN-M	range	ATTN-S	range
rot_mid_identity	identical	0.5025	0.5012-0.5046	0.5041	0.5012-0.5069
rot_flat_identity	identical	0.5008	0.4979-0.5036	0.5000	0.4982-0.5021
rot_mid	rotated, PR 2.65	0.4174	0.4171-0.4180	0.4292	0.4247-0.4320
rot_flat	rotated, PR 3.55	0.4482	0.4477-0.4489	0.4680	0.4660-0.4702

The two *identity* rows are the nulls and both sit at chance, which validates the pipeline. The two rotated rows are the result. All three groups there share one spectrum, so every group is equally hard and no difficulty confound is available — yet AUC sits 0.07 to 0.08 away from chance, and reproduces across independent training seeds to within 0.001. This is a systematic effect, not sampling noise.

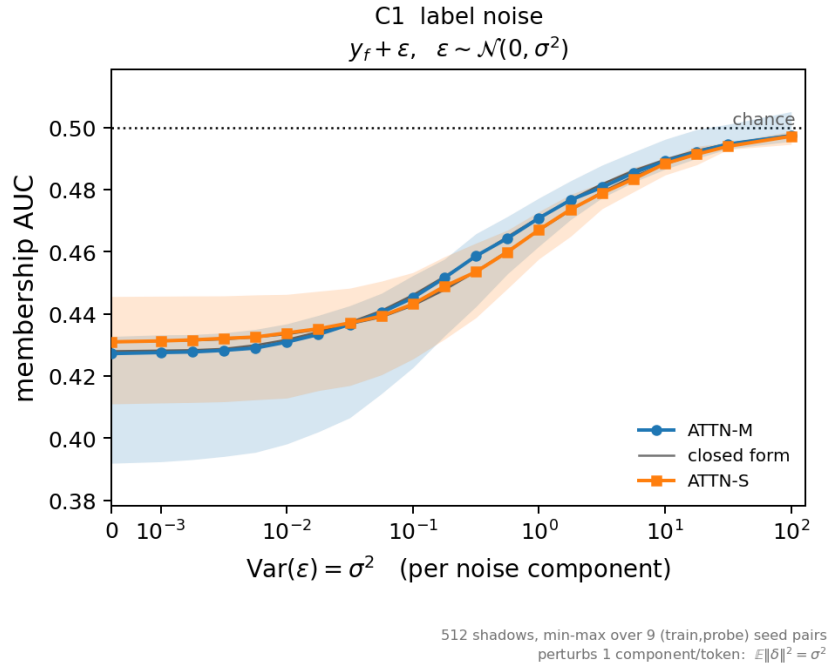
This was predicted to go the other way. The expectation recorded before the run was that equalising the spectra would largely cancel the read-out mismatch between the two hypotheses and push AUC toward 0.5. It does not. The prediction is recorded as failed.

3.2 The closed form is exact

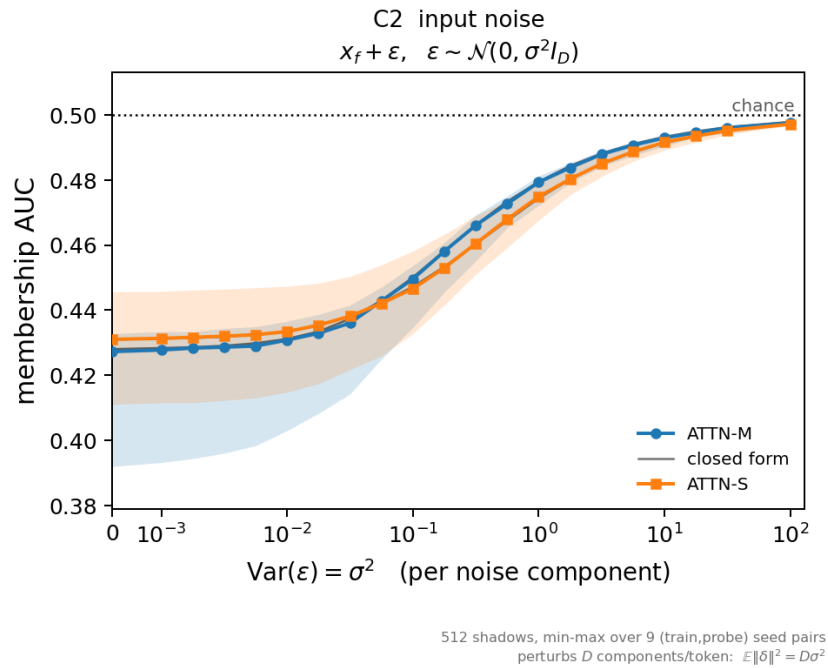
configuration	arch	arm	mean error	max error
rot_mid	ATTN-M	C1	0.00048	0.00176
rot_mid	ATTN-M	C2	0.00039	0.00163
rot_mid	ATTN-S	C1	0.00045	0.00199
rot_mid	ATTN-S	C2	0.00034	0.00140
rot_flat	ATTN-M	C1	0.00040	0.00180
rot_flat	ATTN-M	C2	0.00032	0.00147
rot_flat	ATTN-S	C1	0.00044	0.00188
rot_flat	ATTN-S	C2	0.00036	0.00153

Absolute difference between the analytically predicted AUC and the measured AUC, across the whole corruption grid. Agreement at 4×10^{-4} means the simulation pipeline and the algebra are independently confirming each other; a systematic gap would have indicated a bug in one of them.

3.3 The corruption sweeps



C1, additive label noise, on rot_mid. Both architectures rise monotonically from the baseline toward chance and stop there. The thin dark line is the closed-form prediction; the band is the spread across nine seed combinations.



C2, additive input noise, same configuration. Approaches chance faster than C1 at matched noise variance.

3.4 Neither arm crosses chance

configuration	arch	arm	min AUC	max AUC	crosses 0.5?
rot_mid	ATTN-M	C1	0.4273	0.4974	no
rot_mid	ATTN-M	C2	0.4273	0.4975	no
rot_mid	ATTN-S	C1	0.4310	0.4972	no
rot_mid	ATTN-S	C2	0.4310	0.4971	no
rot_flat	ATTN-M	C1	0.4463	0.4988	no

rot_flat	ATTN-M	C2	0.4463	0.4989	no
rot_flat	ATTN-S	C1	0.4650	0.4972	no
rot_flat	ATTN-S	C2	0.4651	0.4989	no

A second pre-registered prediction, also failed. The label-noise arm carries a mean shift that grows linearly in the noise variance while the spread grows only as its square root, so it was predicted to overshoot chance and keep going. It does not: across four decades the maximum reached is 0.4989. The mean-shift term is real and is included in the closed form, which still matches to 4×10^{-4} , so the algebra is not at fault — the coefficient is simply too small in the trained models to matter over this range.

3.5 Masking versus removal

Percentage of the movement toward chance attributable to variance inflation rather than to information removal, on rot_mid, ATTN-M.

arm	s2=0.01	s2=0.1	s2=1	s2=10	s2=100
C1	116%	109%	66%	17%	1%
C2	99%	102%	94%	67%	22%

Below roughly unit variance both arms are essentially pure masking: the attacker is defeated by added spread, not because the forget information has gone. Genuine removal only takes over at large corruption, and much sooner for the label arm than the input arm. This matters for any claim that a corruption “worked”: at small corruption strengths it has not removed anything.

4. Corrections made along the way

The flip arm did not match the brief. The existing label-flip implementation was deterministic, $y_f \rightarrow (1-2t)y_f$, whereas the brief specifies a Bernoulli(theta) flip. These share a mean, $E[(1-2B)y] = (1-2\theta)y$, but the Bernoulli version also carries variance $4\theta(1-\theta)y^2$. Given that most of the measured effect at small corruption is masking, a corruption family with no variance channel cannot exhibit the dominant mechanism. A correct Bernoulli arm was added; section 5 derives it.

The masking diagnostic's sign convention is inverted here. It is documented as “positive means part of the AUC *drop* is masking”, which assumes AUC starts above chance and falls. Every configuration measured starts *below* chance and rises, so every masking value is negative and the documented reading inverts. The numbers are correct; the label is misleading.

Figure filenames collided across configurations. Two plotting scripts wrote names that omitted the configuration, so a multi-configuration batch silently overwrote its own output. Fixed and covered by a test.

The null-control test was statistically invalid. Its first version required every per-row confidence interval at zero corruption to cover 0.5. That fails on correct data for three compounding reasons: at zero strength all six corruption modes are the identity edit, so one measurement was counted six times; requiring 108 simultaneous 95% intervals all to cover has probability $0.95^{108} = 0.4\%$ under a perfect null; and the nine seed combinations reuse three trained ensembles, so they are not independent. The replacement deduplicates, clusters on the training seed, and adds an absolute bound because a three-seed t-test has almost no power.

5. The Bernoulli arm

With $\tilde{y}_i = (1-2B_i)y_i$ and $B_i \sim \text{Bernoulli}(\theta)$, write $a_i = y_i(c_x \cdot x_i)$ over the forget slice, where c_x is the input block of the read-out covector. Then

$$\begin{aligned} E[\text{delta yhat}] &= -2 \theta / (N+1) \cdot \sum a_i \\ \text{Var}[\text{delta yhat}] &= 4 \theta (1-\theta) / (N+1)^2 \cdot \sum a_i^2 \end{aligned}$$

The step that makes this cleaner than the additive label-noise arm is that $(1-2B)^2 = 1$ identically, so a sign flip leaves the label-label block of the context vector *exactly* unchanged. The quadratic drift that complicates C1 has no analogue here, and the label block of the read-out never enters. Verified at 400,000 trials per point:

theta	E[d yhat] measured	predicted	Var measured	predicted
0.05	-0.001209	-0.001213	3.994e-05	4.002e-05
0.10	-0.002444	-0.002426	7.620e-05	7.582e-05
0.25	-0.006076	-0.006065	1.583e-04	1.580e-04
0.50	-0.012126	-0.012131	2.108e-04	2.106e-04
0.75	-0.018210	-0.018196	1.579e-04	1.580e-04
1.00	-0.024262	-0.024262	0 (exact)	0

The variance is non-monotone in theta: zero at 0, maximal at 0.5, zero again at 1 where the flip becomes deterministic. So the Bernoulli arm and the deterministic arm coincide at theta = 1 and differ most at theta = 0.5.

6. Stage 2: the learned corruption

6.1 The objective minimises the wrong quantity

The brief writes $\min_{\theta} \text{AUC}(\theta) + \lambda \|\theta\|$. Every configuration measured in Stage 1 has AUC *below* 0.5 and rising toward it. Minimising AUC therefore drives the corruption toward greater distinguishability: an AUC of 0.30 is a better attacker than 0.50, merely an inverted one. The success criterion is chance, so the objective must penalise distance from it:

$$\min_{\theta} (\text{AUC}(\theta) - 1/2)^2 + \lambda \|\theta\|$$

Both forms are implemented; the corrected one is the default. This is not a quibble about sign conventions — with the literal objective and a baseline of 0.417, gradient descent moves θ the wrong way.

6.2 The score-function gradient is unnecessary here

The brief derives a REINFORCE estimator because AUC is not differentiable through the sampling of the corrupted labels. That reasoning is sound in general. But for the parameterisation the brief itself specifies — independent per-token Bernoulli flips with features passed through unchanged — the expectation can be taken analytically *before* any sampling. The two moments above are exact for a per-token θ_i , as well as a scalar θ , so this covers both the single-parameter form and the neural policy. Feeding them into the Gaussian AUC makes the objective an ordinary differentiable function.

Measured on a synthetic instance matched to the shipped geometry (D=4, N=31, 11 forget tokens, 64 probe points, 512 shadows), gradient of the corrected objective at $\theta = 0.15$:

estimator	gradient	std. dev.	cost per gradient	noise / signal
closed form	-7.83e-03	0 (exact)	6 ms	0
REINFORCE, 8 samples	-2.65e-02	8.07e-02	43 ms	10.3
REINFORCE, 32 samples	+1.14e-04	4.09e-02	176 ms	5.2
REINFORCE, 128 samples	-8.31e-03	2.04e-02	810 ms	2.6

The score-function estimator is unbiased and does converge on the closed-form value as the sample count grows, which confirms the two are the same objective. But at 128 samples it still costs 135 times more per gradient and its standard deviation is 2.6 times the quantity being estimated.

REINFORCE remains necessary if the policy is extended to perturb the *features*, since it is precisely the $\delta(x_i)$ factor that removes the cross terms. That path is kept in the implementation for that reason.

6.3 The optimal policy, and a limit on the closed form

Optimising the corrected objective drives θ to 0.957, at which point the closed form reports AUC = 0.5000 exactly. The optimum is not a small perturbation: it flips almost every forget label. That is the mean-compensating solution — the flip's mean shift is used to *cancel* the pre-existing membership gap, not to avoid creating one. A corruption constrained to be mean-zero cannot reach chance at all.

The closed form should not be trusted for the last step, however. Measured at the same θ , AUC is 0.514, not 0.500 — a residual gap of 0.014. The reason is that the corruption itself is not Gaussian: the shift is a sum of only eleven Bernoulli terms, and near the optimum that sum has skew +2.3 and excess kurtosis +4.9. This is a genuine difference from the additive-noise arms, where the noise is Gaussian by construction and the same closed form is accurate to 4×10^{-4} .

theta	AUC, closed form	AUC, measured	measured - 0.5
0.000 (baseline)	0.41421	0.41414	0.08586
0.800	0.49100	0.50540	0.00540
0.900	0.49418	0.51328	0.01328
0.957 (closed-form optimum)	0.50000	0.51381	0.01381
1.000	0.51139	0.51157	0.01157

The practical recipe this supports: optimise on the closed form, which is deterministic and roughly a hundred times cheaper per gradient and lands in the right neighbourhood in one shot, then polish against the measured AUC. Do not report the closed-form value as the result for this arm.

7. Status

7.1 Complete

Stage 1 sweeps for all corruption arms, on four control configurations, with nine seed combinations each; the closed-form overlay and its validation; the Bernoulli arm and its Monte-Carlo validation; the Stage 2 policy, both gradient estimators, and the objective correction; and a synthetic-instance demonstration of the whole Stage 2 loop.

7.2 Not yet run

The Stage 2 optimisation has not been run against the *trained* ensembles — only against a synthetic instance built to match their geometry. The script for it is written and reuses the cached ensembles, so it costs seconds of GPU time, not hours. The Bernoulli sweep has likewise not been run on the trained ensembles.

Two questions remain open and both are cheap to settle from the cached ensembles. First, the decomposition of the residual 0.07 baseline signal into read-out mismatch versus genuine memorisation. Second, whether a two-sided attacker — one that detects a *variance* difference rather than a mean difference — still succeeds where the one-sided statistic reports chance. If it does, then “AUC reaches 0.5” overstates what has been achieved, and the success criterion needs strengthening.

7.3 Predictions recorded before the runs

#	prediction	outcome
P8	null controls sit at chance	confirmed
P12	closed form matches measurement	confirmed, 4x10-4
P9	rotation moves AUC toward chance	FAILED
P10	label-noise arm crosses chance	FAILED
P1	arm ratio equals $\text{tr}(\text{inv } \Lambda)/D$	mixed: 4.58 vs 4.55 on rot_flat, off 3.6x on rot_mid
P11	input-noise drop is all masking	true below unit variance, false above

Recording the failures is the point of having written the predictions down first. Two of six were wrong, and one of those — that equalising the spectra would restore chance — is the most consequential result here.

7.4 Files

policy.py (Stage 2 policies and estimators) · theory.py (closed forms, now including the Bernoulli arm) · corrupt.py (corruption families) · stage2_optimise.py (Stage 2 driver) · stage2_poc.py (the synthetic demonstration reported in section 6) · verify_bern.py (Monte-Carlo validation) · check_null.py (the corrected null test) · PREDICTIONS.md (pre-registration)